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MILITARY SPECIFICATION

HEATER, WATER, OIL-FIRED

This limited coordination Military specification has been prepared by U. S. Army Mobility Equipment Command, Directorate of Research, Development and Engineering, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-H-52036A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers an oil-fired, automatically-controlled, water heater capable of raising the temperature of water from 32° F. to 70° F. at the rate of 3,000 gallons of water per hour without contamination or otherwise affecting the potability.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification; Transportation-Data and Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-C-3767/8	- Connector, Plug, Electrical (Power, Four-Wire, Polarized, Twist-Locking Type), Type UP106M.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated (15-Ounce and 21-Ounce Capacity).
MIL-W-6858	- Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-H-15424	- Hand Tools; Packaging of.
MIL-E-16298	- Electric Machines having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-E-55301	- Electromagnetic Compatibility.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS24207	- Coupling, Grease Gun.
MS35844	- Plug, Machine Thread; Magnetic (Drain).

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DRAWINGS

MO(ERDL)

D8259-1 thru -20, -24 thru -29, -31, and -35 thru -38	- Heater, Water, Oil-Fired, 40° F. Temperature Rise at 3,000 GPH.
C13212E9117	- Cover, Electrical Connector.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

ST 1 - Standards Publication for Specialty Transformers.

(Application for copies should be addressed to National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y. 10017.)

AIR MOVING AND CONDITIONING ASSOCIATION, INC. (AMCA)

Bulletin No. 110 - Standards, Definitions, Terms and Test Codes for Centrifugal, Axial and Propeller Fans.

(Application for copies should be addressed to Air Moving and Conditioning Association, Inc. (AMCA), 2159 Guardian Building, Detroit 26, Mich.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, United Engineering Center, 345 East 47th Street, New York, N. Y. 10017.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A233 - Mild Steel Arc-Welding Electrodes.
A371 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.
B285 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AMERICAN IRON AND STEEL INSTITUTE

Steel Product Manual - Stainless and Heat Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, N. Y. 10017.)

3. REQUIREMENTS

3.1 Description. The heater shall be as shown on Drawings D8259-1 through -20, -24 through -29, -31, and -35 through -38 and C13212E9117 and as specified herein. The heater shall effect a temperature rise of not less than 38° F. in water flowing through the heater at the rate of 3,000 gallons per hour. The effluent temperature shall be not less than 70° F. or more than 75° F.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the heater. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 Preproduction model. The supplier shall furnish a heater within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods will produce heaters that

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comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish heaters conforming to this specification.

3.3 Initial production. Unless otherwise specified (see 6.2), the supplier shall furnish to the Government one or more heaters (see 6.2) for testing as specified in 4.4.

3.4 Material. Material shall be as shown on the applicable drawings.

3.5 Electrical components.

3.5.1 Motor. The motor shall be continuous duty, splash proof, design B, 208 volts, 3-phase, 60 cycles, 3,600 rpm synchronous speed. The shaft shall be extended on both ends. The shaft for the pulley shall be 3/4 inch in diameter and 2-5/8 inches long. The shaft for the blower shall be of the diameter and length required by the blower. The direction of rotation facing the blower end shall be counterclockwise.

3.5.2 Cable assembly. The cable assembly shall conform to figures 1 and 2.

3.5.3 Ignition step-up transformer. The ignition step-up transformer shall conform to NEMA Standard ST 1, radio interference suppressed, with the following characteristics:

Primary voltage	208 volts
Secondary open-circuit voltage rating.	10,000 volts
Secondary short-circuit current rating.	23 ma
Input rating at secondary short circuit.	250 va

3.5.4 Thermostatic switches.

3.5.4.1 Effluent water temperature control. The thermostatic switch for control of effluent water shall operate on approximately a 5° F. differential over a water temperature range of 70° F. to 75° F. and shall be initially set at 70° F.

3.5.4.2 High-limit temperature control. The thermostatic switch for high-limit temperature control shall automatically break the circuit when the temperature of the water in the heater rises to 180° F. and shuts down the heater.

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3.5.5 Ignition electrodes. The electrodes shall withstand the heat within the burning chamber and shall operate on 10,000 volts. The probe of the electrode shall be 4 inches long.

3.5.6 Solenoid valve. The solenoid valve shall be of the magnetic-lever-action, 2-wire-control, normally-closed type with packless construction. The valve shall operate on 110 volts, 60 cycles, shall not leak when closed, and shall open against a port pressure of 150 psig. The valve shall be equipped with 1/4 NPT inlet and outlet connections. The direction of flow shall be permanently marked on the outside of the valve.

3.5.7 Low-water electrode. The low-water electrode shall actuate the low-water relay if the water level in the boiler drops below the tip of the electrode. The metal parts of the electrode shall conform to AISI type 304, annealed. The insulating sleeve shall be nylon or polytetrafluoroethylene.

3.5.7.1 Low-water relay. The low-water relay shall cause the combustion controller to shut down the heater if the water level in the heater falls below the probe of the low-water electrode. The relay shall operate on 110 volts, 60 cycles, and shall be a 2-pole type requiring manual reset after each shutdown.

3.5.8 Electrical control components.

3.5.8.1 Combustion controller. The combustion controller shall be an electronic device that shall respond instantly to the actuation of a photo-electric cell assembly. The controller shall operate on 110 volts, 60 cycles. During starting, the controller shall supply power to flame failure indicator light and the programmer. When the photoelectric cell detects the presence of flame, the combustion controller shall complete the circuit to the solenoid valve, thus holding the valve open after the programming controller cuts off the starting circuit to the valve. The combustion controller shall also break the circuit to the indicator light. When the photoelectric cell does not detect the presence of flame during start, the controller shall shut down the heater in 35 to 55 seconds after initial starting. When flame failure occurs during the running cycle, the controller shall revert to its starting position within 0.8 second and attempt to restart the heater. The controller shall require manual resetting when the heater fails to start after a flame failure. The controller shall automatically sequence through the operation of all operating and safety controls and interruption of electric current.

3.5.8.1.1 Photoelectric cell assembly. The photoelectric cell assembly shall register the presence or absence of burner flame and shall actuate the controller under the following conditions: burner ignites; burner fails to ignite during the starting period; flame failure occurs after the starting period; and burner stops and flame is extinguished.

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3.5.8.2 Programmer. The programmer shall consist of a synchronous motor drive, cam and mu-switch, and fuel and ignition time delay relay, all operating on 110 volts, 60 cycles. When power is supplied to the programmer, the programmer shall energize the ignition transformer to provide ignition arc immediately. Within 15 seconds, the programmer shall close the circuit to the solenoid valve and shall keep the solenoid valve circuit closed for 15 seconds and then open it. Two seconds after the solenoid valve circuit is opened, the ignition transformer shall be deenergized and the programmer shall lock. When power from the controller is stopped, the programmer shall immediately reset to starting position.

3.6 Blower. The blower shall have a capacity of not less than 450 cubic feet per minute (cfm) of air under a static pressure not less than 5.4 ounces when the impeller is rotated at not less than 3,450 rpm.

3.7 Fuel pump. The fuel pump shall be a 2-stage, gear-type pump suitable for use on a 2-line system and shall be equipped with a strainer, an air-bleeder valve, and a pressure-adjusting screw for adjusting the delivery pressure over a range of 75 to 150 psi. Interior components exposed to fuel oil shall be corrosion-resisting metal. When operated at a shaft speed of 1,725 rpm and pumping fuel oil conforming to VV-F-800, grade DF-2, the pump shall delivery not less than 20 gallons per hour (gph).

3.8 Oil nozzle tip. The delivery of 8.3 gph at a spray angle of 80 degrees plus or minus 5 degrees shall be at a pressure not exceeding 140 psi with fuel oil conforming to VV-F-800, grade DF-2.

3.9 Electromagnetic compatibility. The electromagnetic compatibility of the heater shall be in accordance with MIL-E-55301, tactical support class.

3.10 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.10.1 Lubricants. The heaters shall operate as specified herein when lubricated with Military lubricants (see 6.4).

3.10.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.10.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

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3.10.4 Filling, draining, and checking provisions. Enclosures such as gear cases that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs shall be used to convey the lubricant from the drain to a point where it can be collected in a container. The drain plug, the filling means, and the lubricant level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.10.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827 as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F.

The heaters shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because Military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which Military grease has been used.

3.10.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of Military oils designated for use in the temperature range and grade of lubricants used.

3.11 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that surfaces are free from all foreign matter which interfere with the adherence or function of the varnish.

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3.12 Identification marking. The heaters shall be identified in accordance with MIL-STD-130. The marking shall be applied to the unit of identification plates conforming to the requirements of MIL-P-514, type I, style 2, photographic plate material.

3.13 Designation plates. All controls, starters, receptacles, lights, thermometers, and switches, including reset switches, shall be designated. Designation shall include nomeclature, and, where applicable, operational characteristics, such as voltage, temperature, pressure, and method of resetting. Marking shall be applied to plates conforming to MIL-P-514, type IX, photographic plate material. Plates shall be attached to the item with rivets. Where attaching by rivets is not practical, plates shall be attached to the item with soft wire. Plates shall be not less than 1-1/4 inches by 3 inches. Marking shall be black printing on a light gray background.

3.14 Caution plates. The switch box and control panel box shall be provided with caution plates indicating voltage. Plates shall conform to MIL-P-514, type IX, photographic plate material and shall be attached to the outside of covers with rivets. Plates shall be not less than 1-1/2 inches by 3 inches. Marking shall be black printing on a yellow background.

3.15 Wiring diagram. A wire diagram reflecting actual wiring layout used shall be located on the inside of the control panel box cover. Drawing D8259-24 shall be used as a guide in preparation of the diagram. The wiring diagram shall conform to MIL-P-514, type IX, photographic plate material. Marking shall be black printing on a white or light gray background.

3.16 Treatment and painting. The portions of the heater not in contact with water shall be cleaned, treated, and painted in accordance with MIL-T-704.

3.17 Technical publications to accompany heaters. Such technical publications as are specified shall be furnished (see 6.2).

3.18 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.19 Items to be furnished with heater. No shipment of heaters shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the heaters without such items has been received from the contracting officer.

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3.20 Workmanship.

3.20.1 Metal fabrication. Metal used in the fabrication of the heaters shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Sheared, sawed, and chipped edges shall be free of burrs and slivers. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing for steel only. Methods utilizing heat shall not be used for nonferrous metals. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Holes shall be punched or drilled and shall have the burrs removed.

3.20.2 Bolted connections. Washers and lockwashers shall be provided and all bolts, nuts, and screws shall be tight.

3.20.3 Riveted connections. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surfaces of the member.

3.20.4 Welders and welding. Unless otherwise specified on the drawings, welders and welding shall conform to the following.

3.20.4.1 Welders. Before assigning any welder to manual welding work covered by this specification and applicable drawings, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed:

Standard Qualification Procedures of the American Welding Society.
Welding Qualification of the American Society of Mechanical Engineers.

3.20.4.2 Weld filler metals. Aluminum filler metal shall conform to ASTM B285, type ER5356 or ER5556, except that filler metal ER4043 may be used in highly restricted locations if conditions cannot be eliminated by adjustments of procedures or fixtures or when joining base metal type 6061 to other types of aluminum. Steel filler metal shall conform to ASTM A233, type E6010 or E6011 for carbon steel, and ASTM A371, type ER308L for corrosion-resisting steel.

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3.20.4.3 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.20.4.4 Aluminum welding. Aluminum welding, except spot and seam welding, shall be done by the metal-inert-gas method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations specified in 3.20.4.2. All aluminum welds shall be made by the automatic-machine method when feasible. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metal shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed, or chipped but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 400° F. for a total time of 30 minutes. No process requiring the use of welding flux shall be used.

3.20.4.5 Steel welding. All steel welding, except spot and seam welding, shall be done by the shielded-electric-arc method in accordance with the applicable requirement of the American Welding Society publication or by any other method providing equivalent mechanical properties obtainable using filler material specified in 3.20.4.2. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. Work shall be positioned for flat welding whenever possible.

3.20.4.6 Spot and seam welding. Spot and seam welding shall be in accordance with MIL-W-6858, class C.

3.20.4.7 Rewelding defects. Portions of welded joints that have been rejected because of defects may be repaired only by rewelding. The defective area shall be removed by chipping or machining. Aluminum shall not be cut by methods utilizing heat. Before being rewelded, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration through the joint.

3.20.5 Threads. Threads shall be in accordance with Handbook H28.

3.20.6 Cleaning. The heater shall be cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the heater unsuitable for its intended use. The heater shall be completely drained before preparation for delivery.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Heaters failing a performance test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The heater shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The heater shall be tested as specified in 4.6.2.1 through 4.6.2.5. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. An initial production heater(s) will be selected at random by the Government from the first five heaters being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production heater shall not exclude the remaining heaters from the quality conformance inspection and acceptance provisions specified in section 4.

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4.4.1 Inspection failure. Failure of an initial production heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production heaters and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on heaters previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all heaters accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. AQL shall be 2.5 percent defective.

4.5.3 Tests. Each heater shall be tested as specified in 4.6.2.6. Failure of the test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The heater shall be examined for the following defects:

101. Dimensions not as shown on the drawings.
102. Assembly not complete or not correct.
103. Parts or components missing.
104. Parts or components do not function.
105. Material not as specified.
106. Treatment and painting not as specified.
107. Lubrication devices not accessible.
108. Workmanship not as specified.
109. Heater not completely drained and cleaned before preparation for delivery.
110. Accessories, identification or other marking, operating and safety instructions, or technical publications missing.
111. Repair parts and maintenance tools missing.

4.6.2 Tests.

4.6.2.1 Blower. The blower shall be tested in accordance with AMCA Bulletin No. 110. A capacity of less than 450 cfm at a static pressure of 5.4 ounces and a speed of 3,450 rpm shall constitute failure.

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4.6.2.2 Fuel pump. The fuel pump shall be operated to deliver oil conforming to VV-F-800. The delivery pressure shall be set first at 75 and then at 150 psig, with the pressure-adjusting screw. The pressure shall be read approximately 1 minute after the adjusting screw has been set. The capacity of the pump shall be measured volumetrically. A pump capacity of less than 20 gph at a shaft speed of 1,725 rpm shall constitute failure.

4.6.2.3 Oil nozzle body. Allow the fuel oil specified herein to spray on a flat surface placed perpendicular to the axis of the nozzle body when that axis is horizontal. The fuel oil shall be pressurized to 100 psig and shall flow at a rate of 10 gph. Measure the distance (x) from the center of the spray pattern to the outer edge of the spray pattern. Measure the distance (y) from the tip of the nozzle to the perpendicular plate. When (x) is divided by (y), a quotient of less than 0.7933 or more than 0.9163 shall constitute failure.

4.6.2.4 Operational tests (6 hours). The operational tests shall be conducted using the fuel specified herein and tap water. The tap water shall be at a temperature below 70° F. The heater shall be started and operated for 6 hours. Fuel shall conform to VV-F-800, grade DF-2.

4.6.2.4.1 Operation. During the 6 hours of operation, the heater shall be checked for flue gas and fuel oil leaks, and the temperature of the effluent water shall be noted. Any flue gas or fuel oil leaks shall constitute failure. A failure of the heater to shut down automatically when the effluent water temperature is more than 75° F. or a failure to restart when the effluent water temperature falls below 70° F. shall constitute failure.

4.6.2.4.2 Flame failure. While the heater is in operation, simulate flame failure and note the operation of the photoelectric cell, the combustion controller, and the programmer. Failure of the photoelectric cell to actuate the combustion controller; failure of the combustion controller to provide power to the flame failure indicator light and the programmer during starting; failure of the controller to shut down the heater after an absence of flame of 35 to 55 seconds duration during restarting; failure of the controller to revert to its starting position within 0.8 second after initial flame failure and failure of the controller to attempt to restart the heater; failure of the programmer to energize the transformer to provide ignition arc immediately when provided with power from the combustion controller; failure of the programmer to lock after ignition; or failure of the programmer to reset to normal starting position when power from the controller is stopped, shall constitute failure of this operation test. At the completion of this phase of operational tests, the combustion controller shall be allowed to cool for 5 minutes before restarting.

4.6.2.4.3 Low-water relay. While the heater is in operation, shut off incoming water, open the drain valve on the heater, and note the operation of the low-water relay. Failure of the low-water relay to shut down the heater when the water falls below the probe of the low-water electrode shall constitute failure.

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4.6.2.4.4 High-limit temperature control. Raise the temperature of the water in the heater to 180° F., and note the operation of high-limit temperature control. Failure of the control to shut down the heater when the temperature is 212° F. shall constitute failure.

4.6.2.4.5 Capacity. During the final 2 hours of the operational test, water shall be supplied to the heater at a rate of not less than 3,000 gph. The influent and effluent temperatures shall be recorded at 15-minute intervals for the 2 hours. A temperature rise of less than 38° F. shall constitute failure of this test. To permit the 38° F. temperature rise, influent water may be cooled or the effluent water temperature control may be reset or bypassed.

4.6.2.5 Electromagnetic compatibility. The preproduction model, equipped for the reduction of radio interference in accordance with 3.9, shall be tested as specified in MIL-E-55301. The supplier shall furnish the contracting officer the reports required by MIL-E-55301, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production heaters.

4.6.2.6 Operational test (3-cycle). The heater shall be operated so as to perform not less than three on-off cycles of effluent water temperature control. Failure of the heater to shut down automatically when effluent water temperature is more than 75° F. or failure to restart when the effluent water temperature falls below 70° F. shall constitute failure of this test. Any flue gas or fuel oil leaks shall also constitute failure of this test.

4.6.2.6.1 Flame failure. Flame failure shall be determined in accordance with 4.6.2.4.2.

4.6.2.6.2 Low-water relay. Low-water relay shall be determined in accordance with 4.6.2.4.3.

4.6.2.6.3 High-limit temperature control. High-limit temperature control shall be determined in accordance with 4.6.2.4.4.

4.6.2.6.4 Capacity. The heater shall be operated with water supplied to the heater at a rate of not less than 3,000 gph nor more than 3,050 gph. A water temperature rise of less than 38° F. shall constitute failure of this test. To permit the 38° F. temperature rise, influent water may be cooled or the effluent water temperature control may be reset or bypassed.

4.7 Inspection comparison. The Government may select heaters at any time during the contract production period and subject these heaters to the examination specified in 4.6.1 and the tests specified in 4.6.2.1 through 4.6.2.4.4 to determine the quality of the selected heaters is equal to the quality standards established during initial production testing. The inspection will be performed by the Government at a site selected by the Government. Heaters will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production heater.

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4.7.1 Inspection failure. Failure of any heater when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of heaters until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Tests. The preproduction pack shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, any loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visual damage to the heater shall constitute failure of the preproduction pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack of any level prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Materials, methods, and containers not as specified for levels A or B.	Each incorrect material, method, or container shall be considered one defect.
113.	Preservative applied to surfaces which will come in contact with water.	
114.	Belts not removed or released from tension for level A or B.	
115.	Electric motors not preserved in accordance with the referenced document for level A or B.	
116.	Gate valves and drain cocks not left open for levels A and B.	
117.	Pump not preserved in accordance with the referenced document for level A or B.	
118.	Miscellaneous equipment requiring protection not preserved and packaged as specified for levels A and B.	

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119. Tools not preserved and packaged in accordance with the referenced document for level A or B.
120. Intermediate packaging not as specified for levels A and B.
121. Components not blocked, braced, anchored, and arranged as specified.
122. Depreservation guide not prepared and attached as specified.
123. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The preproduction pack is a pack which will be identical to the pack which the supplier will subsequently produce in fulfillment of the contract. A preproduction pack shall be made, examined, and subjected to the test specified for the applicable level to determine conformance to preservation, packaging, packing, and marking requirements. Approval of the preproduction pack shall not relieve the supplier of his obligation to preserve, package, pack, and mark the heaters in accordance with this specification. Any changes or deviations of production packs from the approved preproduction pack shall be subject to the approval of the contracting officer.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.2.1 Caution. Contact preservatives shall not be applied to any surface, other than the interior of the pump, which will come in contact with water.

5.2.2 Levels A and B.

5.2.2.1 Preservatives. Surfaces of components and parts of the heater requiring the application of a contact preservative in accordance with MIL-P-116, except as specified in 5.2.1, shall be coated with preservatives conforming to the applicable specifications listed in and shall be applied in accordance with MIL-P-116. Surfaces of bolts, nuts, linkages, pins, grease fittings, levers, adjusting rods, clamps, capscrews, and couplings, or any other surfaces requiring the application of a contact preservative not definitely specified herein shall be coated with type P-1 preservative.

5.2.2.2 Electrical components.

5.2.2.2.1 Photoelectric cell assembly, ignition electrodes, and electrode ignition cables. The photoelectric cell assembly, ignition electrodes, and electrode ignition cables shall be preserved together in accordance with MIL-P-116, method IA. The container shall be a close-fitting paperboard box conforming to PPP-B-566, style, type, and class optional.

5.2.2.2.2 Motors. Electric motors shall be packaged in accordance with level A requirements of MIL-E-16298, by the alternate method as specified therein for assembled machines.

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5.2.2.2.3 Control panel boxes, low-water cut-out electrode assembly, switches, solenoids, starter boxes, and transformers. All openings into the control panel boxes, low-water cut-out electrode assembly, switches, solenoids, starter boxes, and transformers shall be sealed with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3.

5.2.2.2.4 Electric wiring system. Exposed ends of wires and openings in sockets, connectors, terminals, conduit, receptacles, and junction boxes shall be sealed with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3. The electric wiring and the tape shall be coated with electrical-ignition, waterproofing varnish conforming to MIL-V-13811.

5.2.2.2.5 Cable assemblies. Each cable assembly shall be coiled to 10 times its diameter, and the coils shall be secured with equally spaced wraps of cotton tape or twine conforming to PPP-T-60, class 3.

5.2.2.3 Fuel pump and burner. Interior surfaces of the fuel pump and burner head assembly shall be coated with type P-10, grade 2 preservative. The preservative shall be applied to the pump while the pump is being actuated and to the burner head assembly by fogging through the ignition electrode bracket. Openings into the pump and burner shall be sealed with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3.

5.2.2.4 Glass. The glass faces in the rear cover assembly and peekhole assembly shall be covered with greaseproof barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3.

5.2.2.5 Thermometers and pressure gages. All openings into the metal cases shall be sealed with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3. The glass faces shall be covered with greaseproof barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3.

5.2.2.6 Belts and pulleys. Belts shall be released from tension. Unpainted pulleys shall be coated with primer conforming to TT-P-664. Removed belts shall be secured to the pulley for which it is intended.

5.2.2.7 Gaskets. Gaskets exposed by disassembly shall be placed in and afforded protection on the affixed component by covering with greaseproof barrier material conforming to MIL-B-121, type I, grade C, secured in place with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3.

5.2.2.8 Fuel supply tubing. The fuel supply tubing shall be coiled, and the open ends shall be sealed with waterproof, pressure-sensitive, adhesive tape conforming to PPP-T-60, class 3.

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5.2.2.9 Wire rope and clips. Wire rope and rope clips shall be coated with type P-1 preservative. The wire rope shall be coiled, and the rope and clips shall be preserved together in accordance with MIL-P-116, method IC-3.

5.2.2.10 Flue retarders. Flue retarders shall be coated with type P-1 preservative and preserved together in accordance with MIL-P-116, method IC-5.

5.2.2.11 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.2.12 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with MIL-H-15424, level A.

5.2.2.13 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.2.14 Intermediate packaging.

5.2.2.14.1 Repair parts. The repair parts shall be intermediate packaged in boxes conforming to PPP-B-601, domestic type, style optional. The boxes shall be close fitting. Strapping shall not be required.

5.2.2.14.2 Components. Packaged and unpackaged components such as flue retarders and photoelectric assembly shall be intermediate packaged in a container specified in 5.2.2.14.1.

5.2.3 Level C. The water heater, components, repair parts, tools, and publications shall be preserved and packaged to afford protection against corrosion, deterioration, and physical damage during shipment from the source of supply to the first receiving activity.

5.3 Packing.

5.3.1 Level A, B, and C. The heater mounted on its base shall be enclosed with top, sides, and ends of a crate conforming to MIL-C-104, type II, class 2. Sheathing shall be cut to allow sides to fit over lifting flanges mounted on the base, and end sheathing shall extend to the lower edge of the wood end members attached to the base. Loose components and intermediate packages not specifically mentioned shall be secured to the heater or to the base in any available space that will not increase cubage or cause damage.

5.4 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide," and secured in a conspicuous location on the heater. Unless otherwise specified (see 6.2 and 6.5), DA Form 2258, "Depreservation Guide for Engineer Equipment," shall be used.

5.5 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The heater covered by this specification is intended for heating potable water under arctic conditions before the water is stored in tanks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When initial production inspection is not required (see 3.3).
(c) Number of heaters to be furnished for initial production testing when applicable (see 3.3).
(d) Technical publications required (see 3.17).
(e) Applicable list of repair parts and maintenance tools (see 3.18).
(f) Time interval for submission of electromagnetic compatibility reports (4.6.2.5 and 6.3).
(g) Level of preservation and packaging required (see 5.2). Level B preservation is intended to provide economical but limited protection and should be specified only when it is determined that the water purification units will be held in covered storage.
(h) When other than DA Form 2258 is to be used (see 5.3).

6.3 Electromagnetic compatibility. The contracting officer should forward a copy of the test report (see 4.6.2.5) to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, N. J.

6.4 Lubricants. The contracting officer should furnish a list of Military lubricants applicable to the heater covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 DA Form 2258. The contracting officer should arrange to furnish DA Form 2258 when requested by the supplier.

Custodian:

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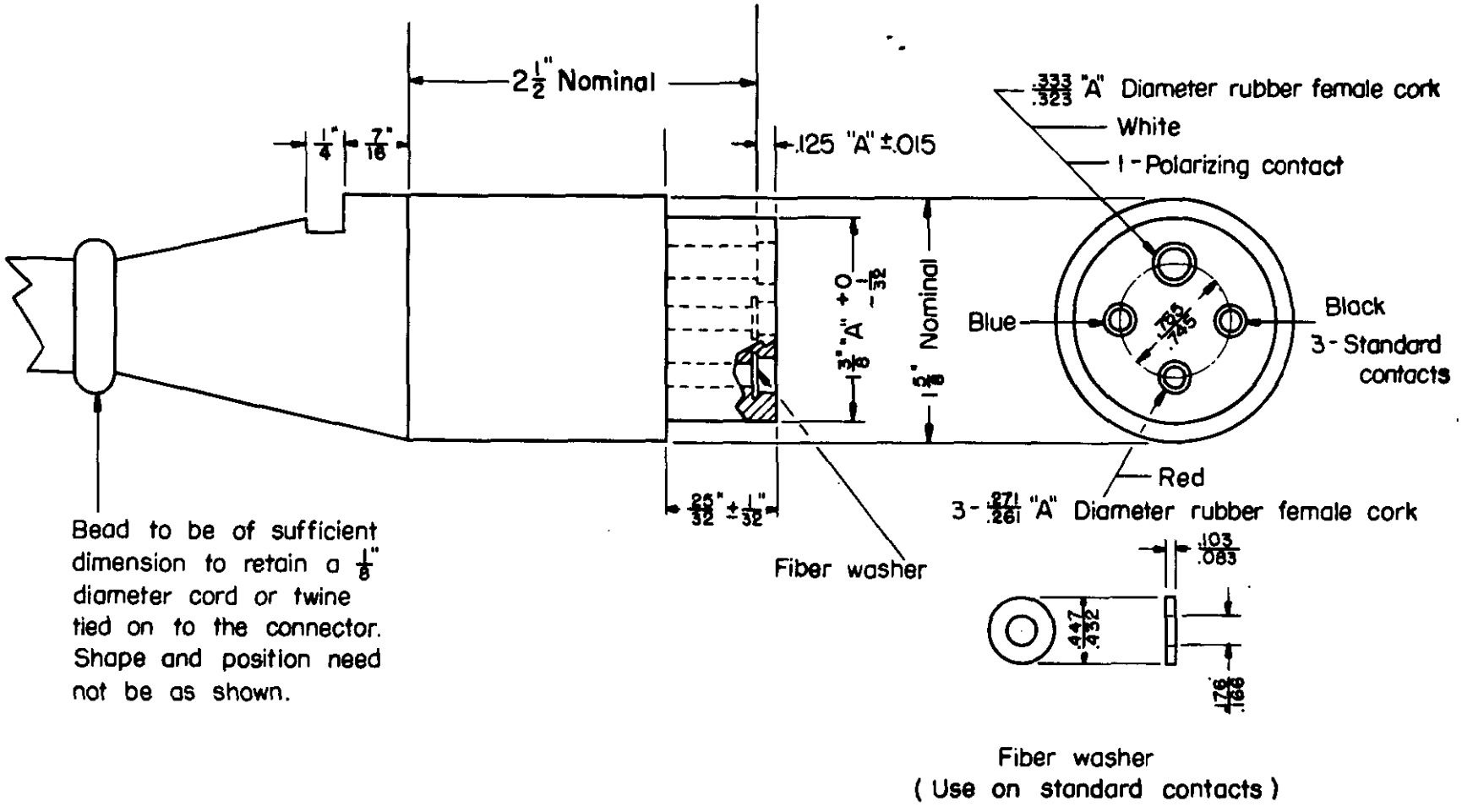
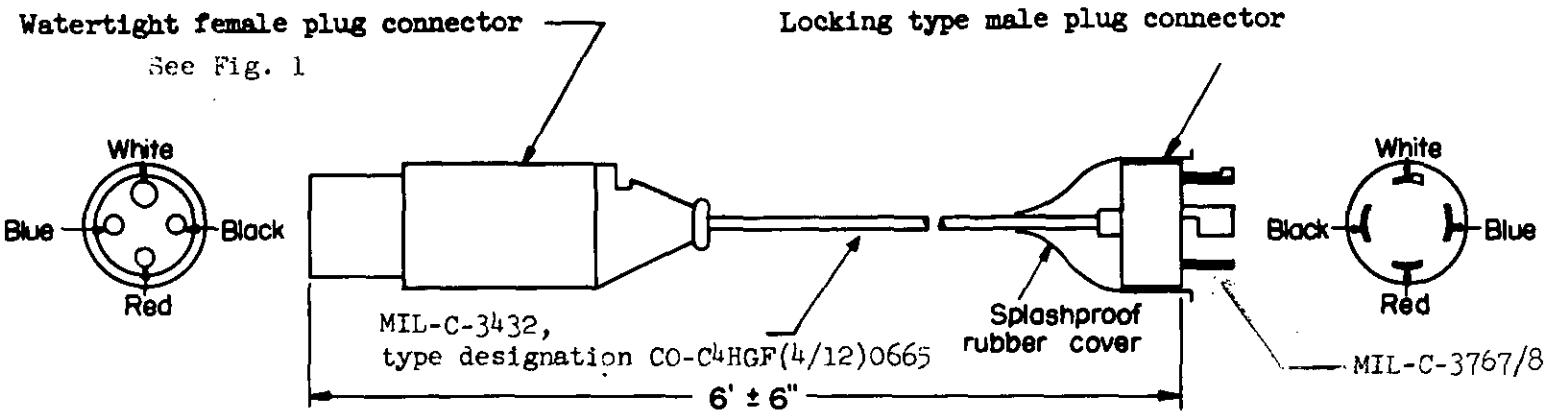


FIGURE 1- WATERTIGHT FEMALE PLUG CONNECTOR

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Note: Molded lock type cap acceptable.
If molded cap is used, a $\frac{1}{4}$" skirt shall be provided for splashproofness.
Color coding shall be as indicated.

FIGURE 2 · CABLE ASSEMBLY, POWER, ELECTRICAL, 4-CONDUCTOR, ONE END W/WATERTIGHT FEMALE PLUG CONNECTOR, OTHER END W/LOCKING TYPE MALE PLUG CONNECTOR

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).*

SPECIFICATION
MIL-H-0052036B(ME) - Heater, Water, Oil-Fired

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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